MIL-P-63448A (AR)
29 February 1984
SUPERSEDING
MIL-P-63448 (AR)
26 May 1981

MILITARY SPECIFICATION

PROJECTILE, 105MM, HEAT-T-MP, M456A2 SPIKE AND CAP ASSEMBLY

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements and provides quality assurance provisions for the fabrication and packing of the spike and cap assembly for HEAT-T-MP, M456A2 Projectile.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Surfaces For Organic Coatings

FSC 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspections By Attributes

MIL-STD-1169 - Packaging, Packing and Marking For Shipment of Inert Ammunitions Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9327312	-	Packing and Marking for Box, Fiberboard, For Spike and Cap Assembly or Spike Assembly M456A2
9323913	-	Spike
9335940	-	Spike and Cap Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

ASTM Methods - American Society for Testing and Materials

E8	-	Tension Testing of Metallic Materials
E10	-	Brinell Hardness of Metallic Materials, Test for
B117	-	Salt Spray (Fog) Testing

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Phila., PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Heat treatment or stress relief. The heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in each component the mechanical properties specified on the applicable component drawing.

3.2.1 Cold shaping (if applicable). Cold shaping shall be so accomplished as to produce the required mechanical properties in the metal parts. After cold work, the metal parts shall be stress relieved at the time and temperature specified on the applicable drawing to achieve stress relief and still meet the required mechanical properties.

3.3 Protective coating. Protective coating shall be in accordance with the applicable drawings and specifications. When protective coating is required by component details, finish requirements will apply to all indicated surfaces of the component regardless of contour configuration. Salt spray testing shall be in accordance with the applicable drawings.

3.4 Painting. Painting shall comply with the requirements on the applicable drawings. All paint and marking shall be dry to the touch before testing and packing for shipment.

3.5 Process description and controls. The contractor in accordance with the contract provisions shall make available a written description of the specified production process and process controls, determined by the Government as crucial that are to be used in the production of the first article sample (see paragraph 4.3.4)

3.6 Structural integrity. There shall be no breakup or deformation of the spike during any proving ground test conducted.

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3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects

3.8.1.1 Components. All components, and component cavities shall be free from cracks, splits, cold shuts, inclusions, pipes, porosity, bursts, folds, seams, and other metal defects.

3.8.1.2 Component cavities. In addition to 3.8.1.1 all component cavities shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits, craters, and sharp edges.

3.8.2 Burr. No component shall have a burr.

3.8.3 Foreign matter. All parts and assemblies shall be free of dirt, chips, oil, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following sample quantities as indicated.

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Part Description	Drawings	Quantity
Spike (Complete including paint)	9323913	35
Spike (complete excluding paint)	9323913	5

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

4.3.4 Process description and controls. At the same time as submission of the first article samples, the contractor shall make available a written description of the production process and controls being used in the manufacture of the item. Written descriptions shall be submitted on the following component, Spike, dwg. 9323913. Included shall be the intermediate steps in the press operation used in forming the spike, as well as intermediate process temperatures and their specific limits.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-63448A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Spike				9323913
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
I.	Spike (Prior to Paint)				
	a. Examination for Defects	5		3.1	4.4.2.1
	1. Hardness test	Note 1		3.1	4.5.1.1
	2. Tension test	Note 1		3.1	4.5.1.2
	3. Sectional inspection	Note 1		3.1	4.5.2
II.	Spike (After Paint)				
	a. Examination for Defects	35		3.1	4.4.2.1 (Note 2)
	b. Ballistic test	25		3.6	4.5.3
NOTE 1.	After the dimensional and visual inspection of the 5 samples is completed, perform hardness test. The high and the low readings of the hardness test shall be used to perform the tension test. The three remaining spikes from the hardness test shall be used for the sectional inspection.				
2.	After the dimensional and visual inspection of the 35 samples is completed, subject five samples to the salt spray test and five to the adhesion of paint test. The remaining 25 shall be subject to the ballistic test.				

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation of MIL-A-48078.

4.4.1.1 Heat treatment lots. Heat treatment lots shall contain spikes from not more than one heat of steel, that has been heat treated in the same heat-treating equipment, at the same temperature, in one unchanged process. One or more batches may be comprised of a heat treatment lot but lot size shall not exceed 2000 components. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.2 Stress relieved lots. Stress relieved lots shall contain spikes from not more than one heat of steel cold shaped by the same process, utilizing the same cold shaping equipment (as applicable) and stress relieved in one unchanged process. One or more batches may be comprised of a stress relieved lot but lot size shall not exceed 2000 components. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.3 Miscellaneous lots. Small lots (such as spikes manufactured from abnormally small heats of steel or spikes that have become separated from previously delivered lots, etc) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1 or 4.4.1.2 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.4 Spike lots. When it has been determined that spikes of a heat treated, stress relieved or miscellaneous lot have complied with the mechanical properties requirement of the applicable drawings, they may be regrouped into spike lots, to which will be assigned a spike lot number. The spike lots shall consist of spikes produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision.

4.4.2 Examination. See MIL-A-48078.

a. Unless otherwise specified in the classification of defects and tests, sampling plans for major and minor defects shall be in accordance with MIL-STD-105 Level II.

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b. The roughness comparison specimens prescribed by ANSI-B 46.1 shall be used as a basis for surface roughness determination.

c. Tool control dimensions. Dimensions marked tool control shall be gaged during first article testing and after each 250 items produced on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor. All material produced prior to periodic inspection will be rejected and screened whenever a defective tool control dimension is found.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 5		DRAWING NUMBER
4.4.2.1	Spike				9323913
					NEXT HIGHER ASSEMBLY
					9335940
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Runout of large inside surface to large outside surface		0.40%	3.1	Gage
102	Runout of O-Ring groove diameter to pitch diameter of shoulder thread		0.40%	3.1	Gage
103	Runout of spike stem to pitch diameter of nose thread, and front face		0.40%	3.1	Gage
104	Runout of spike stem to pitch diameter of rear thread and body seat		0.40%	3.1	Gage
105	Runout of inside diameter of spike stem to pitch diameter of rear thread and body seat (if applicable)		0.40%	3.1	Gage
106	True position of inside diameter of spike stem to major diameter of rear thread (if applicable)		0.40%	3.1	Gage
107	True position of insulator cavity to pitch diameter of nose thread		0.40%	3.1	Gage
108	True position of pitch diameter of collar thread, to pitch diameter of rear thread and body seat		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.1	Spike		2 OF 5	9323913
				NEXT HIGHER ASSEMBLY
				9335940

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	Distance from rear face to forward inside basic diameter		0.40%	3.1	Gage
110	Distance from spike shoulder to second forward outside basic diameter (if applicable)		0.40%	3.1	Gage
111	Distance from spike shoulder to first forward outside basic diameter (if applicable)		0.40%	3.1	Gage
112	Distance from rear face to rear inside basic diameter		0.40%	3.1	Gage
113	Distance from front face to front face of shoulder		0.40%	3.1	Gage
114	Distance from rear face to body seat surface		0.40%	3.1	Gage
115	Distance from body seat surface to front of shoulder		0.40%	3.1	Gage
116	Distance from front face of spike to rear face of nose thread undercut		0.40%	3.1	Gage
117	Distance from forward face of shoulder to rear face of undercut		0.40%	3.1	Gage
118	Pitch diameter of rear thread, min.		0.40%	3.1	Gage
119	Major diameter of rear thread, min.		0.40%	3.1	Gage
120	Pitch diameter of nose thread, min.		0.40%	3.1	Gage
121	Major diameter of nose thread, min.		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 5		DRAWING NUMBER
4.4.2.1	Spike				9323913
					NEXT HIGHER ASSEMBLY
					9335940
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
122	Pitch diameter of collar thread, min.		0.40%	3.1	Gage
123	Major diameter of collar thread, min.		0.40%	3.1	Gage
124	Length of collar thread undercut		0.40%	3.1	Gage
125	Diameter inside spike stem		0.40%	3.1	Gage
126	Diameter rear of insulator cavity		0.40%	3.1	Gage
127	Diameter of collar thread lead-in		0.40%	3.1	Gage
128	Large inside diameter		0.40%	3.1	Gage
129	Depth of insulator cavity		100%	3.1	Gage
130	Thickness at juncture of spike stem and shoulder, min. (see note 1)		0.40%	3.1	Gage
131	Outside radius at juncture of spike stem and shoulder		0.40%	3.1	Gage
132	Parallelism of front face with body seat surface		0.40%	3.1	Gage
133	Body seat gap		0.40%	3.1	Gage (See Note 15 Dwg 9323913)
134	Weight incorrect		0.40%	3.1	Approved scale
135	Parallelism of front face of shoulder with body seat surface		0.40%	3.1	Gage
136	Outside radius at juncture of spike stem and shoulder not tangent		0.40%	3.1	Visual (See 6.5)
137	Threads damaged or malformed		0.40%	3.1	Visual
138	Metal defective		0.40%	3.8,1	Visual
139	Flatness of rear face of insulator cavity		0.40%	3.1	Gage

NOTES:

1. The thickness shall be gaged at the location indicated in the drawing.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Spike	4 OF 5	9323913
			NEXT HIGHER ASSEMBLY 9335940

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
140	Thickness of paint		0.40%	3.1	Gage
141	Outside diameter of spike at second forward basic length from shoulder (if applicable)		0.40%	3.1	Gage
142	Outside diameter of spike at first forward basic length from shoulder (if applicable)		0.40%	3.1	Gage
143	Large outside diameter		0.40%	3.1	Gage
144	Total length, max.		0.40%	3.1	Gage
145	Runout of large outside diameter to pitch diameter of rear threads		0.65%	3.1	Gage
	stem and rear cavity		0.40%	3.1	Gage
146	Salt spray	See Note 2		3.3	See Note 2
147	Adhesion of paint	See Note 2		3.3	See Note 2
148	Hardness test	See 4.4.3.1.1		3.1	4.5.1.1
149	Tension test	See 4.4.3.1.2		3.1	4.5.1.2
150	Sectional inspection	See 4.4.3.2		3.1	4.5.2
MINOR					
201	Diameter of insulator cavity		0.65%	3.1	Gage
202	Diameter of rear thread undercut		0.65%	3.1	Gage
203	Diameter of O-Ring groove		0.65%	3.1	Gage
204	Diameter of collar thread undercut		0.65%	3.1	Gage

NOTES: 2. The salt spray test (use ASTM B117 to perform this test) and the adhesion of paint test shall be conducted in accordance with TT-C-490. The actual painted spike shall be used in these tests. The sampling plan and the rejection criteria is defined in TT-C-490.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Spike	5 OF 5	9323913
			NEXT HIGHER ASSEMBLY: 9335940

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
205	Diameter of nose thread undercut		0.65%	3.1	Gage
206	Width of wire slot		0.65%	3.1	Gage
207	Length of rear thread undercut		0.65%	3.1	Gage
208	Width of O-Ring groove		0.65%	3.1	Gage
209	Length of nose thread undercut		0.65%	3.1	Gage
210	Depth of O-Ring groove		0.65%	3.1	Gage
211	Length of forward shoulder thread lead-in		0.65%	3.1	Gage
212	Distance across insulator cavity and wire slot		0.65%	3.1	Gage
213	Pitch diameter of nose thread, max.		0.65%	3.1	Gage
214	Angle of O-Ring groove wall, excessive		0.65%	3.1	Visual (see 6.1)
215	Radii or chamfer missing or incorrect		0.65%	3.1	Visual (see 6.1)
216	Surface finish improper		0.65%	3.1	Visual (see 6.1)
217	Foreign matter		0.65%	3.8.3	Visual
218	Pitch diameter of shoulder thread, max.		0.65%	3.1	Gage
219	Stamp missing, incorrect or illegible		0.65%	3.1	Visual
220	Burr		0.65%	3.8.2	Visual
221	Paint on threads, body seat and shoulder area		0.65%	3.1	Visual
222	Angle at forward inside basic diameter		Note 3	3.1	Visual (note 4)
223	Protective coating inadequate or damaged		0.65%	3.1	Visual
224	Pitch diameter of collar thread, max.		0.65%	3.1	Visual

NOTES:
3. Samples which are to be inspected will be determined by paragraph 4.4.3.2.
4. See paragraph 4.5.2.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Spike and Cap	1 OF 1	9335940
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
101	Cap improperly assembled		0.65%	3.1	Visual
102	Cap loose or damaged		0.65%	3.1	Visual/Manual

NOTES:

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4.4.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.5.

4.4.3.1 Mechanical properties

4.4.3.1.1 Hardness

4.4.3.1.1.1 Heat treatment lots. In order to determine uniformity of heat treatment, each spike in the heat treatment lot (as defined in 4.4.1.1) shall be subjected to a hardness test with a Brinell hardness tester, using a 5 second minimum dwell or by any other method of hardness testing approved by the technical activity. In any event, details regarding the detailed test and calibrating procedures shall be submitted to the technical agency for approval.

4.4.3.1.1.2 Stress relieved lots. Twenty-five cold shaped spikes (as applicable) from each stress relieved lot (as defined in 4.4.1.2) representing the first product through the stress relief furnace and five spikes from each one hour production thereafter for the balance of the stress relieved lot shall be subjected to a hardness test to assess uniformity of the product. In any event, details regarding the detailed test and calibrating procedures shall be submitted to the technical agency for approval.

4.4.3.1.1.3 Miscellaneous lots. Miscellaneous heat treatment lots of spikes shall be hardness tested in accordance with 4.4.3.1.1.1, miscellaneous cold shaping lots of spikes shall be hardness tested in accordance with 4.4.3.1.1.2.

4.4.3.1.2 Tension test

4.4.3.1.2.1 Heat treatment lots. Two spikes (as applicable) shall be selected from each heat treatment lot, one of the hardest and one of the softest as determined by 4.4.3.1.1.1 shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of one or more tension test specimens to comply shall be cause for rejection of the lot of components represented by the failing specimen. On tangentially transverse specimens the elongation shall be recorded for information only. At the option of the contractor, a hardness range may be used to screen heat treatment lots with spikes falling outside the established range being rejected. Prior to implementing the use of a hardness range, data and methods used to establish the hardness range must be submitted to the technical agency. Spikes which fall outside of the established range may be re-heat treated to fall within the range. Regardless of whether a hardness range is used or not, the high and low hardness spikes from each lot shall be subjected to

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tension testing. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot represented.

4.4.3.1.2.2 Stress relieved lots. Two spikes, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five spikes, shall be subjected to this test. On tangentially transverse specimens the elongation shall be recorded for information only. When it has been determined that the mechanical properties comply, the furnace temperature and time will be established and this temperature and time shall be maintained throughout each stress relief lot. The minimum allowable furnace temperature shall be 700 degrees fahrenheit with a minimum soak time of 30 minutes. Furnace temperature and time will be recorded every hour to assess adequate control of the stress relief equipment. The accuracy of temperature measuring and controlling instruments shall be checked each eight hours by an independent method. Failure of any tension test specimen to comply with mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot.

4.4.3.1.2.3 Miscellaneous. Samples for tension test of miscellaneous heat treatment or stress relieved lot of spikes shall be selected and tested in accordance with 4.4.3.1.2.1 or 4.4.3.1.2.2 as applicable. Failure of one or more tension test specimens to comply shall be cause for rejection of the lot of components represented by the failing specimens. On tangentially transverse specimens the elongations shall be for information only.

4.4.3.2 Spike sectional inspection. To determine conformance of the spike, after final finishing two (2) samples shall be inspected from each shifts production as per 4.5.2. The samples shall be taken one from the middle of the work shift and one from the end of the work shift. An additional sample shall be taken, if any change to the production process is made, directly after the change is made. Failure of any sample to comply with the requirements of the applicable drawing and specification referenced thereon shall be cause for rejection of the product produced subsequent to the last inspection.

4.4.3.3 First article ballistic test (excess pressure). The twenty-five samples listed in Table I shall be assembled into complete rounds and be ballistically tested to determine compliance with para. 3.6. Failure of any sample to comply with 3.6 shall be cause for rejection of the first article. See para. 4.5.3.

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4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1 The contractor shall submit for approval inspection equipment design in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and Section 6.3 herein.

4.5 Test method and procedures

4.5.1 Mechanical properties. The minimum mechanical properties requirements shall apply throughout the component unless otherwise designated. The number of specimens and their location are for the convenience of the contractor and government and shall not restrain the contractor or government from testing additional specimens at other locations to determine that properties of the components conform to drawing requirements.

4.5.1.1 Hardness. The Brinell hardness test shall be made on the large outside diameter of the spike. A reading shall be taken at one location. Prior to performing the test, the area in which the reading is to be taken shall be cleaned of scale and other foreign matter by grinding a flat of nominal depth 0.015 inches on the spike surface. The test shall be performed using equipment and procedure prescribed by ASTM Method E10.

4.5.1.2 Tension test From the sample provided by 4.4.3.1.2.1, 4.4.3.1.2.2, or 4.4.3.1.2.3, two (2) longitudinally tension test specimens conforming the largest obtainable round specimen prescribed by ASTM Method E8 shall be taken 180° apart from the stem portion of each spike (see Figure 1). In addition, two (2) transverse tension test specimens conforming as closely as possible to the largest obtainable round specimen shall be taken 180° apart in a tangentially transverse direction at the midpoint of the flange area of each spike (see Figure 1). Yield strength shall be determined by the offset method using a limiting set of 0.1 percent offset. The tension test shall be performed utilizing equipment and procedures prescribed by ASTM Method E8.

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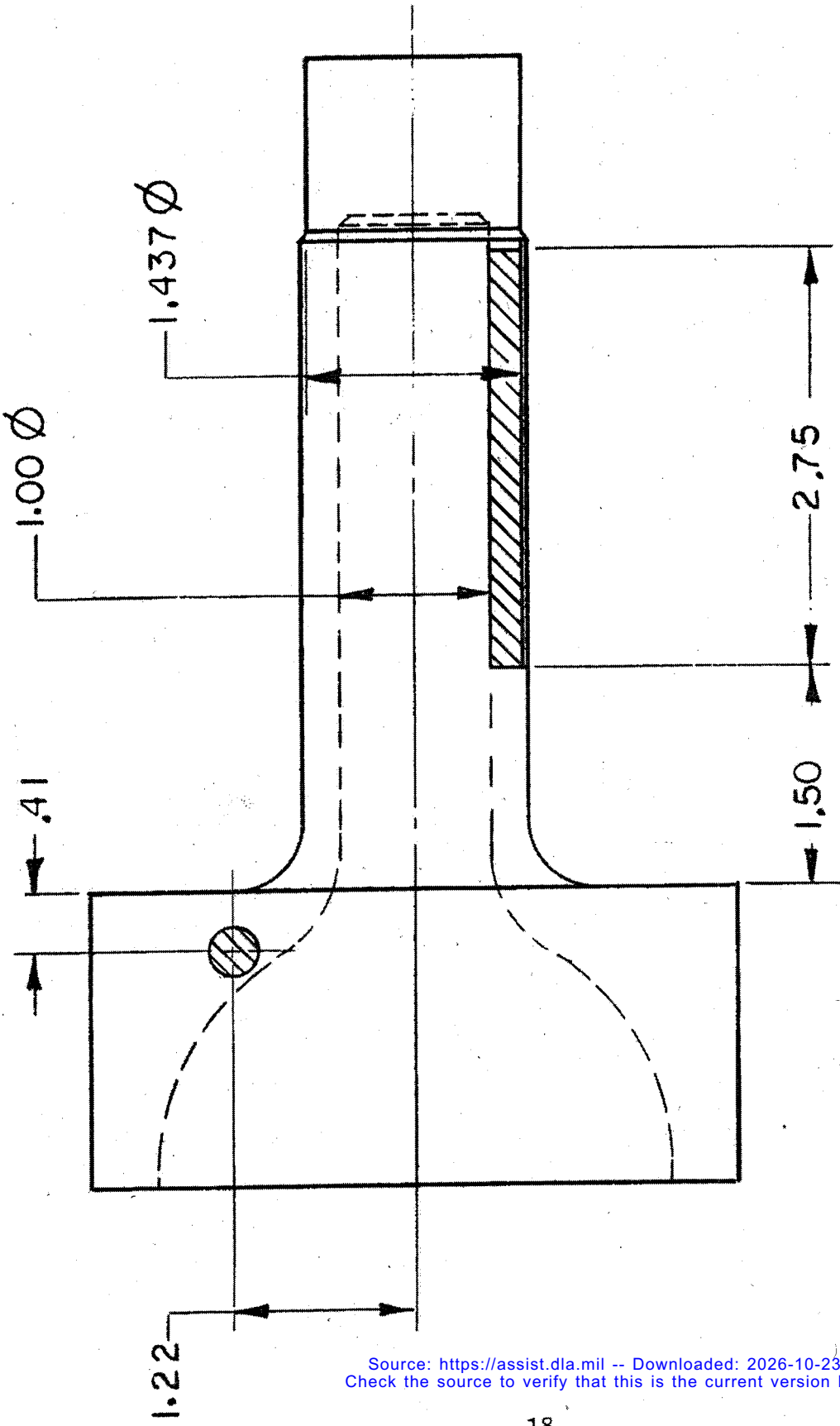


FIG 1.A

LOCATIONS OF TENSILE TEST SPECIMENS

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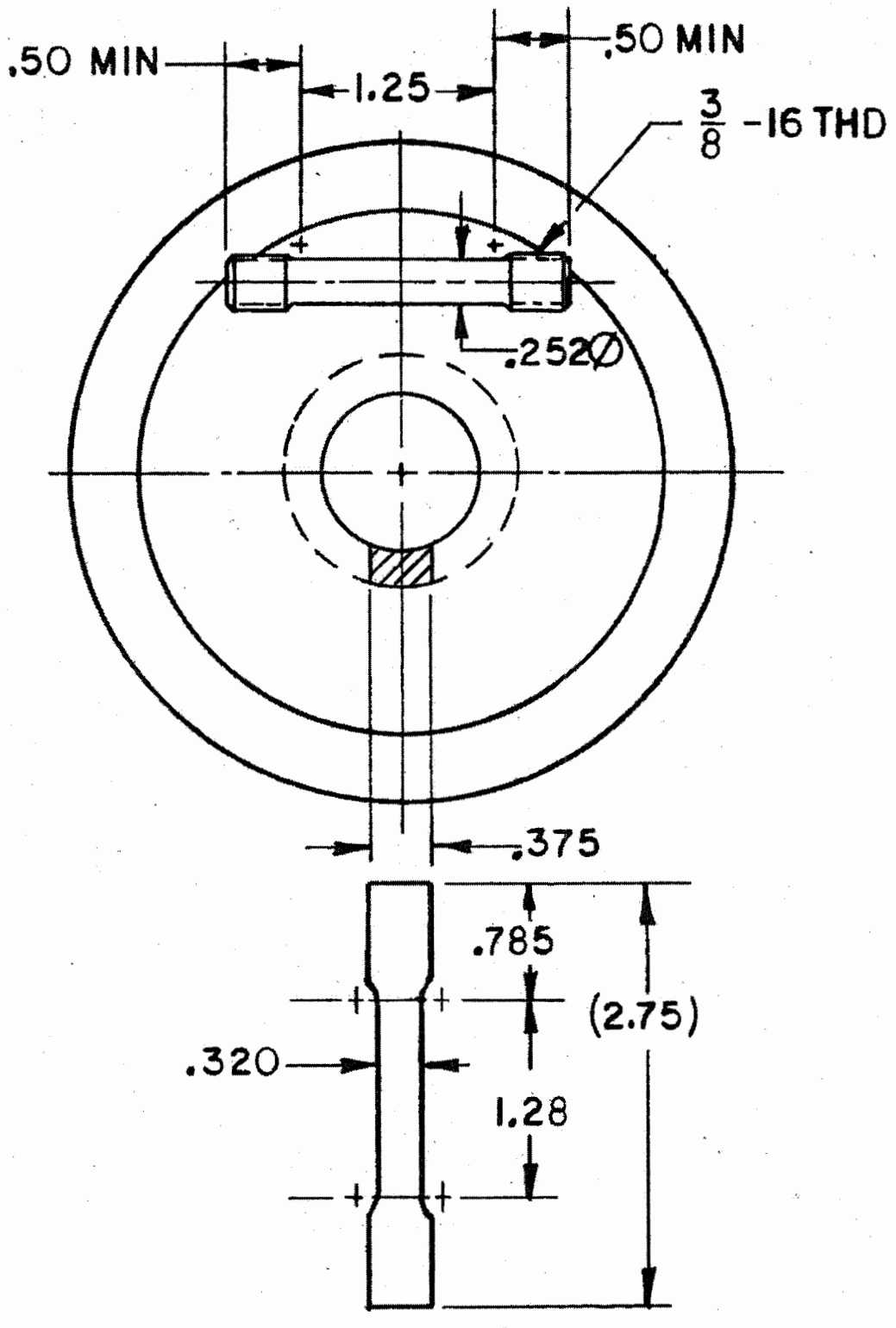


FIG 1.B

SIZE OF TENSILE TEST SPECIMENS

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4.5.2 Sectional Inspection. The samples taken in accordance with 4.4.3.2 shall be cut in half along a center line of the spike. Surface finish shall be determined by a visual inspection and at the same time blend of contour and angle at forward inside basic diameter (evidence of steps not permitted) shall also be examined.

4.5.3 Ballistic test. The first article ballistic test shall be conducted at a government proving ground in accordance with the applicable acceptance test procedure. The sample cartridges shall be inert loaded and fired from a M60 tank series. Photographic equipment will be set up near the muzzle in such a manner as to record the integrity of the spike. The propellant charge will be adjusted to obtain a chamber pressure of 70,000 psi (482.6 MPa) to 73,900 (509.5 MPa). The cartridges shall be temperature conditioned at 70 $\pm$ 2.5°F (21 $\pm$ 1.5°C) for a minimum of 24 hours.

5. PACKAGING.

5.1 Preservation and packing

5.1.1 Level B minimum military pack. The spike and cap assembly for projectile, 105MM, HEAT-T-MP, M456A2 shall be packed in accordance with drawing 9327312.

5.1.1.1 Percentage of extra parts packing. Packing for the required percentage of extra parts shall be provided for by furnishing the proper number of prime containers.

5.1.1.2 Shipping. If the component parts from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated.

5.1.1.3 Marking. The carton shall be marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.2 Ordering data. See MIL-A-48078.

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6.3 Submission of inspection equipment designs for approval. See MIL-A-48078.

Commander
US Army Armament Research and Development Center
ATTN: DRSMC-QAT-I(D)
Dover, N.J. 07801

6.4 Submission of test data and ammunition data cards. In addition to the normal distribution of records when the spike is procured by the Army Armament, Munitions and Chemical Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, US Army ARDC, ATTN: DRSMC-LCU-CV(D) and DRSMC-QAT-A(D), Dover, NJ 07801.

6.5 Visual standards Visual standards are to be established at the contractor's plant for compliance with 3.8.

Custodian:
AMRY-AR

Preparing Activity:
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